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Close Encounters of the AI Kind: Use of AI Influencers As Brand Endorsers

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ABSTRACT

Brand endorsers can contribute to a brand's success or failure (in the case of endorser transgressions). Recent advancements in technology have produced new, nonhuman alternatives to traditional celebrity endorsers. These new endorsers rely on artificial intelligence (AI) to interact with and influence consumers. Two studies demonstrate that AI influencers can produce positive brand benefits similar to those produced by human celebrity endorsers. Moreover, just like their human counterparts, AI influencers can also commit transgressions that result in degradation of the endorsed brand. Importantly, though, AI influencers differ from human celebrity endorsers in that consumers are less likely to view them as unique entities (as tested in a pilot study). Thus, consumers are more likely to perceive a transgression committed by an AI influencer as behavior applicable to all AI influencers, but they are less likely to view celebrity endorser behaviors as interchangeable. As such, after an AI influencer has committed a transgression, replacing the AI influencer with a celebrity endorser attenuates negative brand perceptions, an effect which cannot be realized if the replacement is another AI influencer.

Today's consumers have been described as "always on and constantly connected" (Stephen 2017), resulting in real-time information exchanges with others that are "becoming an integral element of consumer behavior" (Hennig-Thurau et al. 2010). Social media platforms allow for a "more diverse range of voices" (Hines 2019), with both individual consumers and paid endorsers adding to the conversation about various brands. But are all of those *human* voices? In the age of artificial intelligence (AI), the answer is likely to be no. "AI refers to the broad idea that computers, through the use of software and algorithms, can think and perform tasks like humans" (Kumar et al. 2019, p. 135). Davenport et al. (2020) suggest that, as existing technologies are perfected and new technologies emerge, AI will "change the face of marketing strategies and customers' behaviors" (p. 25). Although Davenport and colleagues used the term in a more general sense, in this article we take a more literal approach and specifically consider the face of marketing strategies (i.e., the endorser) and the impact of AI.

A recent analysis of Twitter posts found that about 30% of user-generated content is already created by bots using AI technologies to pose as humans (Liu 2019). Marketers can incorporate AI into their endorsement strategies in various ways. In addition to bots, which often have no visual body representation at all, and spokescharacters, which are obviously animated representations not easily mistaken for human, marketers are also beginning to utilize virtual endorsers. These endorsers have distinct humanlike visual representations and use both technology and human oversight in their interactions with others.

A virtual endorser can have a sizable social network of followers and can be regarded as "a trusted tastemaker in one or several niches" (De Veirman, Cauberghe, and Hudders 2017, p. 798). Los Angeles startup Brud manages three popular virtual endorsers: Lil Miquela, Bermuda, and Blawko, which are ranked first, fifth, and sixth, respectively, on HypeAuditor's 2019 Top Instagram Virtual Influencers list (Baklanov 2019). Lil Miquela, currently described in her

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Instagram profile as a "change seeking robot," is perhaps the most recognizable. In 2018, she was named one of *TIME* magazine's most influential people on the Internet and was described as a style icon promoting luxury brands such as Prada, Balenciaga, and Kenzo (Time 2018). Bermuda, a self-proclaimed "Robot Queen," and Blawko, one of the few male virtual endorsers, also endorse products, but they do so for more mainstream brands such as Netflix, Nike, and Del Taco (Baklanov 2019).

Virtual endorsers may seem attractive to marketers due to perceptions of lower costs or reduced likelihood that the endorser will be involved in a scandal. However, these benefits, like the endorsers themselves, may be illusions. According to HypeAuditor, virtual endorsers often cost more than human endorsers with the same number of followers (Baklanov 2019) and there is some evidence that virtual endorsers, like those run by Brud, commit transgressions just like their human counterparts. For example, initial press coverage of Bermuda was somewhat negative because of a series of disparaging remarks about Lil Miquela, which eventually culminated in Bermuda hacking her rival's Instagram account to expose her as a fellow nonhuman (Koebler 2018). Only later was it revealed to be a staged fight, as both virtual endorsers were created by the same company (Raphael 2018). Critics have also disparaged Blawko for his drug references and disrespectful language (Raphael 2018), and a 2019 Calvin Klein ad featuring Lil Miquela making out with human model Bella Hadid was criticized as queerbaiting (Petrarca 2019).

Even given the controversies, technology-enabled interactions between brands and consumers seem to be a key component for marketers going forward. In fact, one industry insider estimated that in the coming years "50% of all content will be developed by machines, with oversight and editing by humans" (Pophal 2017, p. 7). As technology continues to advance, however, the use of virtual endorsers may soon be replaced by AI influencers (Appel et al. 2020). Indeed, another industry executive suggests that "in the era of influencer marketing, the ability to automatically sense and respond to customer interactions will enable companies with [AI] capability to scale exponentially" (Zhou 2018, p. 10). As such, the focus of this research is on AI influencers. By integrating definitions on AI (Kumar et al. 2019) and influencers (Carrillat and Ilicic 2019), we formally define an AI influencer as a digitally created artificial human who is associated with Internet fame and uses software and algorithms to perform tasks like humans.

Given that the use of AI influencers in marketing is relatively new, little research exists on the implications of using AI influencers; nor is research available to help guide brands when AI controversies arise. The crux of the present research is to fill these gaps. As such, this research answers calls (e.g., Appel et al. 2020; Carrillat and Ilicic 2019; Voorveld 2019) for further examination of social influencers and responds to Keel and Nataraajan (2012), who suggest that research is needed on the types of celebrity endorsers, including characters "who do not have a real, personal life" (p. 698). Thus, this research enhances our understanding of endorser types, specifically AI influencers, while extending previous work on AI in marketing. To date, research on AI in marketing has largely focused on how technology can be utilized to create customized advertising copy and product recommendations (Chen et al. 2019; Deng et al. 2019; Qin and Jiang 2019; Wang et al. 2019), optimize prices, and provide customer service (Rai 2020). Further, this research responds to Carrillat and Ilicic (2019) and Swaminathan et al. (2020), who encourage further investigation into the restorative actions that brands should take in crises resulting from the behavior of brand partners (including endorsers).

This article proceeds as follows. First, we provide an overview of the technologies that allow for the creation of AI influencers. Next, we review the literature on endorsers and suggest that brands may expect similar outcomes in terms of consumer attitudes and intentions regardless of whether an endorser is human or AI. Study 1 then demonstrates that transgressions by AI influencers parallel results of celebrity endorser transgressions, decreasing both perceptions of the endorser and the brand. While Study 1 demonstrates similarities in consumers' perceptions of AI influencers and celebrities, Study 2 demonstrates that consumers do not always view these two endorser types comparably. Study 2 investigates how a brand could effectively disassociate itself from a transgression committed by an AI influencer. Results suggest that replacing an AI influencer who has committed a transgression with a celebrity endorser, as compared to an alternate AI influencer, positively impacts consumer attitudes and purchase intentions toward the sponsoring brand. Brands switching from an AI influencer to a celebrity endorser are seen as acting more responsibly than those that simply replace one AI influencer with another. Managerially, these results provide marketers with a critical succession plan in the case of an AI influencer transgression. The results also impact theory by highlighting potential

differences in the way in which consumers view celebrity endorsers and AI influencers.

AI Influencers

The technologies needed to create AI influencers are already being used in advertising. Kietzmann, Paschen, and Treen (2018) identify five AI building blocks (natural language processing, image recognition, speech recognition, problem solving, and machine learning) that, when used in conjunction, create an AI influencer. The first three are used to gather information. Natural language processing derives meaning from the language contained in consumer texts such social media posts, direct messages, or tweets (Kietzmann, Paschen, and Treen 2018). Image recognition "helps advertisers understand pictures and videos that people share on social media and that 'show' true consumer behavior" (Kietzmann, Paschen, and Treen 2018, p. 264). The information gleaned from the images that consumers post where a brand is visible (even if it is not tagged or mentioned by name) can be used to assess sentiment around a brand. This technology is also able to analyze facial expressions and may in fact be better at doing so than humans (Hutson 2018). Finally, speech recognition is currently used in call centers to better understand customer needs but could also be used by an AI influencer to gather information from videos that consumers post to social media. According to Kietzmann and colleagues (2018), "Personality characteristics that are important in people's lives eventually become a part of their language. The way that AI can 'reason' with people's social media comments and posts, in addition, may reveal personality tendencies, values, and needs" (p. 264). Thus, an AI influencer can become attuned to followers' personalities and subsequently use this information to better inform the ways in which to interact with these followers.

Once gathered, insights from natural language processing, image recognition, and speech recognition can be used by the marketer to problem solve. For example, natural language generation technologies can take the structured data that come out of natural language processing and use them to create new unstructured text or message content (Del Rowe 2019). In the case of an AI influencer this might mean analyzing consumer posts about a brand and then creating content that touches on the same topics. One of the benefits of natural language generation is the ability to maintain a consistency in brand voice (Del Rowe 2019), which may be difficult to achieve when using human influencers. Another advantage is that AI can facilitate both convergent problem solving, wherein a single best solution is sought, and divergent problem solving, which allows a variety of possible solutions to be identified (Paschen, Pitt, and Kietzmann 2020). In practice, AI influencers might use problem solving to build several different posts to respond to consumer sentiment. The AI influencer then has the ability to "self-optimize in real time as a result of AI assessment of behavior and reactions to a multitude of different variations" (Campbell et al. 2020, p. 239).

Machine learning, Kietzmann and colleagues' (2018) final AI building block, allows for continuous improvement in the process and movement toward "fine-grained and insightful" (p. 264) interactions, as the AI influencer learns from every contact with consumers. The use of AI in influencer marketing, therefore, represents a potential lucrative investment for many marketers as it allows brands to have "meaningful, one-on-one conversations with consumers along their paths to purchase" (Campbell et al. 2020, p. 238).

To understand how an endorsement made by an AI influencer may affect a consumer's evaluation of and intentions toward the endorsed brand, we first examine the effectiveness of the more traditionally employed human endorsers. Historically, celebrities—known for their accomplishments in the sports or entertainment world—have been selected as endorsers. However, with the current proliferation of social media, a new type of endorser, the influencer, has emerged. Although some research suggests that celebrities are a type of influencer (Kim and Kim 2020), in alignment with other researchers (e.g., Appel et al. 2020; Breves et al. 2019; Carrillat and Ilicic 2019) we acknowledge that although there are similarities between celebrity endorsers and influencers, influencers and celebrities are distinct endorser types. As Carrillat and Ilicic (2019) note, both influencers and celebrities have social capital, the key difference is that for the celebrity endorser, "status is attained independently of endorsement activities" (p. 64), while influencers use products and brands to aid in the establishment of their social capital. As such, to achieve social capital and endorsement deals, influencers brand themselves as experts on various social media platforms (Appel et al. 2020; Schouten, Janssen, and Verspaget 2020). Although some research suggests influencers may be more effective than celebrities under certain conditions (Schouten, Janssen, and Verspaget 2020), research consistently finds that both types of endorsers provide brands with many benefits.

Past research demonstrates the effectiveness of celebrity endorsers and explores boundary conditions for positive outcomes related to consumer attitudes, intentions, and behaviors (for detailed reviews, see Bergkvist and Zhou 2016; Erdogan 1999; Keel and Nataraajan 2012). Recent research on influencers suggests similar boundary conditions and positive brand outcomes (Breves et al. 2019; Colliander and Dahlén 2011; Kim and Kim 2020; Schouten, Janssen, and Verspaget 2020).

As AI influencers possess social capital, serve as exemplars of taste, have attributes admired by consumers (McQuarrie, Miller, and Phillips 2013), and behave in ways similar to influencers and celebrity endorsers, we anticipate that AI influencers who endorse brands will also generate similar, positive effects for the endorsed brands consistent with these endorser types. Further, research on AI and machine learning also points to potential positive effects that could be realized through the use of AI influencers. For example, Ferrara and colleagues (2016) suggest that the boundary between humanlike and botlike behavior is becoming less distinct, which makes it possible for a bot to acquire significant influence. Research in computer sciences explores the use of bots to spread misinformation and influence crucial events, such as elections (Broniatowski et al. 2018; Ferrara et al. 2016; Shao et al. 2018). In terms of their role in marketing, Liu (2019) finds that social bots (software agents that automatically write and respond to messages on social media) are already effective at influencing brand virality.

Moreover, the premise that AI influencers can positively affect perceptions of endorsed brands is supported by the associative network model of memory, which proposes that information stored in memory is connected through paths, allowing for the activation from one node to the next (Anderson 1983). This model has been used to explain consumers' brand associations (Keller 1993), specifically in the context of endorsers, suggesting that, over time, brands and their endorsers become linked in consumers' memories (Keller 1993; Thomas and Fowler 2016; Till and Shimp 1998; Till 1998). Thus, AI influencers who endorse a brand will become associated with that brand in the mind of the consumer, suggesting that positively perceived AI influencers should result in more positive brand perceptions.

Just as positive perceptions regarding an AI influencer should work in the brand's favor, negative perceptions developed when an influencer behaves badly should result in a decrease in consumer attitudes and intentions toward the brand. Indeed, past research finds that when celebrity endorsers engage in behaviors that are perceived poorly by consumers, the brand with which the endorser is linked suffers. Research shows that after an endorser has committed a transgression, consumers' evaluations of the brand (Fong and Wyer 2012; Till and Shimp 1998; Um 2013) and purchase intentions (Um 2013) decrease, ultimately resulting in a decline in the firm's financial performance (Bartz, Molchanov, and Stork 2013; Louie, Kulik, and Jacobson 2001). Moreover, Carrillat and colleagues (2014) find that not only does the endorsed brand suffer in terms of consumer evaluations after an endorser transgression, other brands in that product category are also evaluated more negatively. Taken together, these results suggest that actions taken by an endorser have the potential to affect consumers' attitudes and intentions toward the endorsed brand. Similarly, we propose that when an AI influencer commits a transgression, the linkage between the AI influencer and the brand (Anderson 1983; Keller 1993) will result in reduced attitudes toward the brand and purchase intentions.

H1: A transgression committed by an AI influencer will negatively impact consumers' (a) attitudes and (b) purchase intentions toward the endorsed brand.

Brand Response to AI Influencer Transgressions

If an AI influencer's transgression can negatively affect the endorsed brand, as proposed by hypothesis 1, marketers need to be prepared with strategies to mitigate such damage to their brand. Past research finds that repentance, whereby the offending party makes "an attempt to accept crisis responsibility and to find remedy" (Singh, Crisafulli, and Quamina 2020), is an effective strategy when a transgression occurs (Dirks et al. 2011; Dutta and Pullig 2011). Given that the actions of the endorser are often not in the control of the brand, the endorsed brand may opt to take matters into its own hands and repent its ill-fated connection to the endorser. This, too, will attenuate the negative evaluations typically levied on brands affiliated with the transgressor (Thomas and Fowler 2016). In this research, we examine a brand's repentance strategy whereby the brand finds remedy by replacing the AI influencer with a new endorser. Hock and Raithel (2020) find stock markets respond positively to brands that have suspended an endorser who has committed a transgression and that such announcements are received favorably by the markets when the endorser is not apologetic. This

suggests that replacing an offending AI influencer may be an effective strategy.

A question arises, though: With whom should the brand replace the AI influencer—another AI influencer or a celebrity endorser? We focus specifically on celebrities, as their use as endorsers is more prevalent than ever (Carrillat and Ilicic 2019) and the implications of celebrity transgressions on endorsed brands is well documented (e.g., Bartz, Molchanov, and Stork 2013; Fong and Wyer 2012; Louie, Kulik, and Jacobson 2001; Till and Shimp 1998; Um 2013). We posit that replacing an AI influencer with a celebrity endorser will help mitigate the negative effects of the AI influencer's transgression, but these same positive outcomes will not be extended to a brand if the AI influencer is replaced with another AI influencer. While individuals are open to technology use, research demonstrates that once an individual has observed a technology fail, even if overall the technology is superior to humans, individuals are less likely to use the technology. Indeed, the technology acceptance model (TAM) finds that if a technological application is unreliable, failing to demonstrate results and usefulness, consumers' intentions to use that technology are reduced (Venkatesh and Davis 2000). Further, within the context of algorithms, Dietvorst and colleagues (2015) find that individuals who see an algorithm err prefer a human forecaster to the algorithm even when the human forecaster is less accurate. Finally, Wirtz and colleagues (2018) suggest consumers have an aversion to machine learning and "quickly lose trust in AI" (p. 918), while Schmitt (2020) goes one step further by proposing that the reaction may be due to speciesism, a bias to treat the human species preferentially over all nonhumans, including robots and other AI technology. Together, this suggests that a transgression committed by one AI influencer will lead to a higher aversion to a replacement AI influencer, thus failing to alleviate the negative attitudes and intentions toward the brand generated by the initial AI influencer's transgression.

To confirm that AI influencers would be seen as interchangeable while celebrity endorsers would be seen as independent, a pilot study was conducted. Participants ($N = 53$: 69% male, average age 37) were recruited and compensated via Amazon.com's Mechanical Turk (MTurk). Participants were asked to complete two sets of questions: one pertaining to AI influencers and one pertaining to celebrity endorsers. The presentation order of the sets (AI influencer or celebrity) was randomized. In each set, participants were first provided a definition of the focal endorser type (AI influencer or celebrity) and then asked to complete three items regarding endorser similarity (adapted from Pappu and Cornwell 2014; Walsh and Mitchell 2005) and two items assessing probability of transgression occurrence with new endorser. Thus, for endorser similarity, participants were asked to rate their agreement with the statements: "All celebrities (AI influencers) are the same"; "Most celebrities (AI influencers) are very similar, making it difficult to distinguish them"; and "Because of the great similarity between celebrities (AI influencers), it would be difficult to detect when a brand gets a new celebrity endorser (AI influencer)." To assess transgression occurrence with a new endorser, participants read,

> Now, please take a moment to imagine that a celebrity endorser (AI influencer) makes a politically incorrect statement. A politically incorrect statement is one that offends or upsets people in society and often deals with attitudes toward sex, race, or disability. If the brand replaced the celebrity endorser (AI influencer) with a new celebrity endorser (AI influencer), what is the likelihood that the NEW celebrity endorser (AI influencer) would engage in the same behavior?

Details on all items used in the studies are provided in Table 1.

A paired-samples t test revealed that participants perceived AI influencers to be more similar/interchangeable ($M = 4.18$, $SD = 1.96$) than celebrity endorsers ($M = 3.71$, $SD = 1.96$; t (52) = 1.96, p = .05). Importantly, a paired-samples t test also revealed that participants perceived a new AI influencer to be more likely to commit a similar transgression to a past AI influencer ($M = 4.40$, $SD = 1.59$) as compared to a new celebrity in the same situation ($M = 3.59$, $SD = 1.96$; t (52) = 3.38, $p < .01$). Taken together, these results suggest that participants are more likely to perceive AI influencers as similar entities prone to making the same mistakes, whereas celebrities are more readily viewed as unique individuals who are not as likely to commit the same transgressions as another celebrity endorser.

The results of this pilot study may have meaningful implications for brands. If the brand fails to replace an AI influencer with an endorser perceived as independent from the first endorser, consumers may continue to fault the brand for failing to take responsibility and appropriate action. Indeed, as the associative network model of memory demonstrates, brands and endorsers become linked (Keller 1993), reinforcing the belief that others are guilty by association (Cleeren, van Heerde, and Dekimpe 2013). Bolstering support for this contention, past research demonstrates that even when a transgression is committed by a single party, those

Table 1. Measures Studies 1 and 2.

Construct	Scale	Item(s)
Pilot Study		
AI influencer similarity ($\alpha = .87$)	7-point Likert-type (1 = *Strongly disagree*, 7 = *Strongly agree*)	• All AI influencers are the same. • Most AI influencers are very similar, making it difficult to distinguish them. • Because of the great similarity between AI influencers, it would be difficult to detect when a brand gets a new AI influencer.
Celebrity endorser similarity ($\alpha = .94$)	7-point Likert-type (1 = *Strongly disagree*, 7 = *Strongly agree*)	• All celebrities are the same. • Most celebrities are very similar, making it difficult to distinguish them. • Because of the great similarity between celebrities, it would be difficult to detect when a brand gets a new celebrity endorser.
New AI transgression likelihood ($r = .83$)	7-point semantic differential	What is the likelihood that the new endorser would engage in the same behavior? *Very unlikely/Very likely* *Definitely would not/Definitely would*
New celebrity transgression likelihood ($r = .76$)	7-point semantic differential	What is the likelihood that the new endorser would engage in the same behavior? *Very unlikely/Very likely* *Definitely would not/Definitely would*
Study 1		
Attitudes toward the brand ($\alpha = .98$)	7-point semantic differential	The brand is . . . *Bad/Good* *Unappealing/Appealing* *Negative/Positive* *Unfavorable/Favorable*
Purchase intentions ($r = .93$)	7-point semantic differential	Please rate your likelihood of purchasing the brand. *Unlikely/Likely* *Definitely would not/Definitely would*
Attitudes toward the endorser ($\alpha = .98$)	7-point semantic differential	Please rate your overall attitudes toward the endorser. *Bad/Good* *Dislike/Like* *Negative/Positive* *Unfavorable/Favorable*
Transgression appropriateness ($r = .49$)	7-point semantic differential	The endorser making politically incorrect statements is: *Unacceptable/Acceptable* *Very severe behavior/Not at all severe behavior*
Study 2		
Attitudes toward the brand ($\alpha = .97$)	Same as Study 1	Same as Study 1
Purchase intentions ($r = .91$)	Same as Study 1	Same as Study 1
Perceptions of responsibility	7-point Likert-type (1 = *Strongly disagree*, 7 = *Strongly agree*)	The brand acted responsibly.

affiliated through brand partnerships or an endorsement deal are still perceived as responsible (Hock and Raithel 2020; Koschate-Fischer et al. 2019; Thomas and Fowler 2016). This suggests that the effect of a brand's endorser replacement strategy following a transgression on attitudes and intentions toward the brand will be mediated by perceptions of the brand acting responsibly.

H2: Replacing an AI influencer who has committed a transgression with a celebrity endorser, as compared to an AI influencer, will positively impact consumers' (a) attitudes and (b) purchase intentions toward the endorsed brand.

H3: After an AI influencer transgression, the relationship between replacement endorser type and (a) attitudes and (b) purchases intentions toward the endorsed brand will be mediated by perceptions of responsibility.

Study 1

Study 1 serves to test the initial premise that AI influencers and celebrity endorsers are perceived similarly and positively by consumers and tests the hypothesis that when an AI influencer commits a transgression, attitudes and purchase intentions toward the sponsoring brand decrease (hypothesis 1). The study employs a 2 (transgression: present, absent) × 2 (endorser: AI influencer, celebrity) between-subjects design with two dependent variables—attitudes toward the brand and purchase intentions—measured.

Procedure

Using MTurk, 184 individuals (56% male, average age 36) were recruited and compensated. After electing to participate, they opened a link to an online survey that was administered through Qualtrics and had to be completed in one sitting.

Consistent with past transgression manipulations (e.g., Carrillat, d'Astous, and Christianis 2014; Carrillat, d'Astous, and Lazure 2013; Thomas and Fowler 2016), participants were provided with a fictitious article announcing an endorsement deal for a brand. A fictitious brand was employed as it can eliminate problems that may arise with real brands, such as preexisting attitudes and familiarity (Geuens and De Pelsmacker 2017; Nelson, Yaros, and Keum 2006), and is consistent with past research (e.g., Sharma and Varki 2018; Thomas and Fowler 2016). Further, the product category, sunglasses, and brand name, Burlay, have both been employed in previous research (Kennedy, Baxter, and Ilicic 2019).

Participants were randomly assigned to a transgression and an endorser condition, both of which were manipulated within the article. Participants assigned to the no-transgression condition read about the endorsement deal with no mention of negative behavior from the endorser, while those in the transgression condition read that the endorser had tweeted a politically incorrect statement. This transgression described is similar to those for which celebrity endorsers have made headline news (Henderson 2018; Todd 2018). Participants were also randomly assigned to an endorser condition. Those individuals in the AI influencer condition read that the deal was with a new AI influencer, while those in the celebrity condition read that the endorsement deal was with an up-and-coming celebrity endorser. See the Online Appendix for the stimuli employed in all studies.

After reading the article, participants were presented with the survey. For the dependent variables, participants were asked to rate their attitudes toward the brand and purchase intentions (Till and Busler 2000). To test the effectiveness of the transgression manipulation, attitudes toward the endorser were measured. Further, to demonstrate that a politically incorrect statement from an endorser is perceived as inappropriate, participants in the transgression condition were asked to rate the extent to which the politically incorrect statement made by the endorser featured in the article was acceptable (Carrillat, d'Astous, and Christianis 2014). Table 1 provides all measures used in the studies.

Results

A manipulation check was conducted to ensure that the transgression manipulation was effective. A two-way analysis of variance (ANOVA) revealed that, as desired, there were significant differences in attitudes toward the endorser based on transgression condition (F (1, 177) = 80.90, $p < .01$), with participants in the transgression condition rating their attitudes toward the endorser as significantly more negative ($M = 2.64$, $SD = 1.73$) than those in the no-transgression condition ($M = 4.76$, $SD = 1.39$). Importantly, there was no main effect of endorser type on attitudes toward the endorser ($p = .83$); nor was there an effect of the interaction between the transgression and endorser manipulation on attitudes toward the endorser ($p = .32$). To further confirm the transgression manipulation, a one-sample t test demonstrated that participants in the transgression condition perceived the statements made by the endorser as inappropriate ($M = 2.91$, $SD = 1.54$) with such perceptions significantly below the midpoint of the scale (midpoint: 4.0; t (83) = −6.46, $p < .01$). Moreover, to confirm that perceptions of appropriateness did not vary by endorser condition, a one-way ANOVA shows that, regardless of endorser condition, there was no difference in perceptions of appropriateness ($p = .29$)

To test hypothesis 1(a), a two-way ANOVA was conducted to examine the effect of the transgression and endorser conditions on attitudes toward the brand. No effects occurred for endorser type ($p = .84$) or the interaction between endorser and transgression ($p = .71$). A main effect of transgression on attitudes toward the brand did emerge (F (1, 180) = 7.21, $p < .01$; see Figure 1). Participants in the no-transgression condition reported significantly higher attitudes toward the brand ($M = 5.33$, $SD = 1.32$) than those in the transgression condition ($M = 4.79$, $SD = 1.34$).

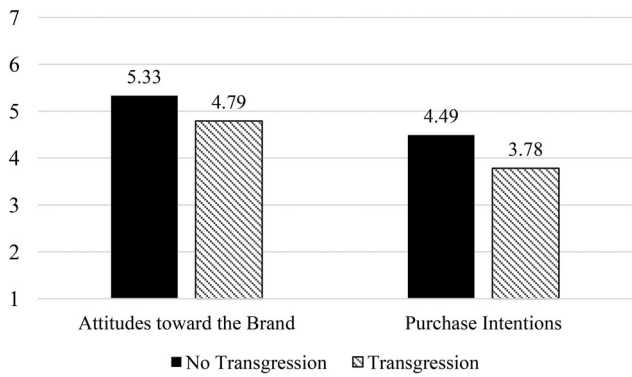


Figure 1. Effect of transgression on attitudes toward the brand and purchase intentions.

Similarly, a two-way ANOVA was conducted to examine the effect of the transgression and endorser conditions on purchase intentions. No effects emerged for endorser type ($p = .83$) or the interaction between endorser and transgression ($p = .32$). The transgression manipulation did have a significant effect on purchase intentions (F (1, 180) = 7.80, $p < .01$) such that participants in the no-transgression condition reported significantly higher purchase intentions ($M = 4.49$, $SD = 1.61$) than those in the transgression condition ($M = 3.78$, $SD = 1.80$; see Figure 1). These results support hypothesis 1(b).

Discussion

The results demonstrate that, regardless of whether an endorser is an AI influencer or a celebrity, when the endorser commits a transgression it has a negative effect on the brand, reducing both attitudes toward the brand and purchase intentions. Thus, these results support hypothesis 1, demonstrating that when an AI influencer commits a transgression that is unrelated to the brand, the endorsed brand still suffers. This account is consistent with the associative network model (Anderson 1983; Keller 1993) and extends existing literature to demonstrate that behaviors by AI influencers have comparable effects on endorsed brands.

In addition to testing hypothesis 1 and providing support that AI influencers' transgressions negatively affect the brand, this study also provides support for the premise that AI influencers are initially perceived similarly to celebrity endorsers. Indeed, a series of one-way ANOVAs demonstrate no significant differences between an AI influencer and a celebrity endorser in the no transgression condition (attitudes toward the brand: $p = .68$; purchase intentions: $p = .58$). Moreover, one-sample t tests demonstrate that when no transgression is present, AI influencers are perceived positively, as attitudes and purchases intentions (attitudes toward the brand: $M = 5.38$, $SD = 1.42$; purchase intentions: $M = 4.62$, $SD = 1.70$) were significantly above the midpoint of the scale (midpoint: 4.0; attitudes toward the brand: t (49) = 6.87, $p < .01$; purchase intentions: t (49) = 2.59, $p < .05$).

Study 2

The goal of Study 2 was to determine how a brand could effectively disassociate itself from a transgression committed by an AI influencer. Past research suggests potential ways to overcome a negative endorser association, with most research examining whether (e.g., Carrillat, d'Astous, and Lazure 2013; Louie and Obermiller 2002) or how quickly (e.g., Vredenburg and Giroux 2018) the break with a celebrity endorser should be announced. To determine an appropriate response when a transgression is committed by an AI influencer, Study 2 was conducted. This study examines consumers' attitudes and intentions toward the brand after an AI influencer who has committed a transgression is replaced with either another AI influencer or a celebrity endorser. The design employs 3 conditions (replacement: AI influencer, celebrity, no replacement/control) and measures attitudes toward the brand and purchase intentions as the dependent variables and perceptions of the brand's responsibility as the mediator.

Procedure

Similar to the previous studies, participants were recruited and compensated through MTurk ($N = 157$: 58% male, average age 37) and completed the online survey through Qualtrics. Twelve participants were removed from the sample for failing to correctly answer the attention check, which required them to correctly indicate that the original endorser was AI. All participants were first provided with the same fictitious article announcing an endorsement deal between a fictitious brand of sunglasses and an AI influencer who had just committed a transgression. Thus, all participants viewed the article from the AI influencer/transgression condition in Study 1. Participants were then randomly assigned to an endorser replacement condition (replacement: AI influencer, celebrity endorser, no replacement/control). Participants in the control condition went on to complete the survey, while those in the AI influencer or celebrity endorser replacement conditions read a brief follow-up news article announcing that the brand was replacing its AI influencer. In the celebrity replacement condition, participants read that the brand was replacing its AI influencer with a celebrity, while in the AI replacement condition, participants read that the brand was replacing its AI influencer with another AI influencer. Consistent with previous research (Carrillat, d'Astous, and Christianis 2014; Thomas and Fowler 2016), all articles were dated so as to provide a sequence of events for the participants. After reading the assigned articles, participants were asked to complete the same dependent measures from Study 1, as well as respond to perceptions of the brand's responsibility (Radighieri et al. 2014).

Table 2. Study 2: Means and standard deviations.

	Celebrity Replacement	AI Replacement	No Replacement
Attitudes toward the brand	5.42_a (1.23)	4.61_b (1.59)	4.75_b (1.48)
Purchase intentions	4.57_a (1.74)	3.62_b (1.75)	$4.05_{a,b}$ (1.84)
Perceptions of responsibility	5.52_a (1.30)	4.96_b (1.46)	4.53_b (1.49)

Note. Different subscript letters indicate mean scores that significantly differ across conditions.

Results

Dependent Variables

A one-way ANOVA shows a significant effect of the endorser replacement condition on attitudes toward the brand (F (2, 154) = 4.51, $p < .01$). Post hoc analysis shows that participants in the celebrity replacement condition reported significantly higher attitudes toward the brand ($M = 5.42$, $SD = 1.23$) than those in the AI replacement condition ($M = 4.61$, $SD = 1.59$; $p < .01$), supporting hypothesis 2(a). Likewise, participants in the celebrity replacement condition reported significantly higher attitudes toward the brand as compared to those participants in the control condition ($M = 4.75$, $SD = 1.48$; $p < .05$). No differences emerged between participants in the AI replacement and control conditions ($p = .61$) (see Table 2 for descriptive statistics).

Similarly, a one-way ANOVA was conducted to examine the effect of the replacement endorser condition on purchase intentions. The replacement manipulation had a significant effect on purchase intentions (F (2, 154) = 3.52, $p < .05$). Providing support for hypothesis 2(b), post hoc analysis demonstrates that participants in the celebrity replacement condition reported significantly higher purchase intentions ($M = 4.57$, $SD = 1.74$) than those in the AI replacement condition ($M = 3.62$, $SD = 1.75$; $p < .01$). No significant differences arose between the control condition and the celebrity replacement or AI replacement conditions ($p = .13$, $p = .22$, respectively) (see Figure 2).

Perceptions of Responsibility

A one-way ANOVA shows a significant effect of the endorser replacement condition on perceptions of responsibility (F (2, 154) = 6.47, $p < .01$). Post hoc analysis shows that participants in the celebrity replacement condition were more likely to perceive that the brand acted responsibly ($M = 5.52$, $SD = 1.30$) than those in the AI replacement condition ($M = 4.96$, $SD = 1.46$; $p < .05$) and those in the control condition ($M = 4.53$, $SD = 1.49$; $p < .01$). No differences emerged between participants in the AI replacement and control conditions ($p = .13$).

Mediation Analysis

Next, data were analyzed using PROCESS Model 4 (Hayes 2013) with 5,000 bootstrap resamples assessing perceptions of responsibility as the mediator, affecting the relationship between the replacement condition and attitudes toward the brand. As the replacement condition is a multicategorical variable, this model was run twice using indicator coding (Hayes and Preacher 2014) to compare the relationship for all three conditions. Results support mediation for the comparison between the celebrity and AI replacement conditions (relative indirect effect: $b = -.37$, 95% confidence interval [CI] = [−.7326, −.0129]) and the celebrity replacement and control conditions (relative indirect effect: $b = -.64$, 95% CI = [−1.018, −.2889]). Consistent with these results, the mediation model was not supported for the comparison between the AI replacement and control conditions (relative indirect effect: $b = .28$, 95% CI = [−.0817, .6711]).

Running the same analyses with purchase intentions as the dependent variable supports mediation for the comparison between the celebrity and AI replacement conditions (relative indirect effect: $b = -.31$, 95% CI = [−.6377, −.0118]) and the celebrity replacement and control conditions (relative indirect effect: $b = -.54$, 95% CI = [−.8890, −.2363]). The mediation model was not supported for the comparison between the AI replacement and control conditions (relative indirect effect: $b = .23$, 95% CI = [−.0808, .5622]).

Discussion

The results support hypotheses 2 and 3. When an AI influencer commits a transgression, replacing the AI influencer with a celebrity endorser results in more positive attitudes toward the brand and higher purchase intentions, relationships which are then mediated by perceptions of responsibility. This bolsters the argument that AI influencers, unlike celebrities, are perceived as less distinct and more interchangeable. Further, the brand's culpability for the endorser's transgression is reduced as perceptions of responsibility are weakened. As no significant differences arose between the control condition and the AI influencer replacement strategy, this suggests that participants

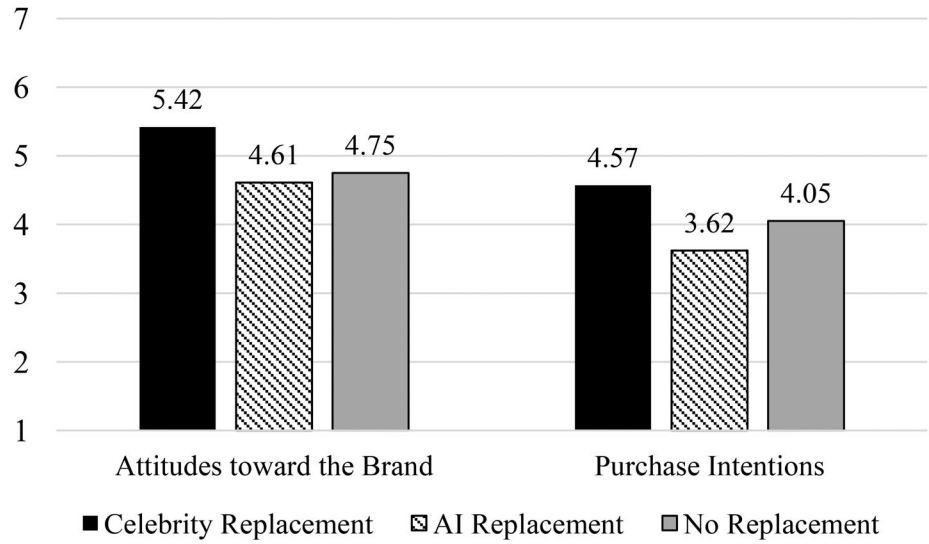


Figure 2. Effect of endorser replacement on attitudes toward the brand and purchase intentions.

failed to view this as an appropriate form of repentance and still held the brand responsible.

General Discussion

Completely autonomous AI influencers are not yet commonplace; most virtual endorsers (such as Lil Miquela, Bermuda, and Blawko) still have some level of behind-the-scenes human intervention. However, Appel and colleagues (2020) predict that completely autonomous AI influencers will soon become more prominent on social media because they will be able to "represent and act on brand values and engage with followers anytime" (p. 83), unlike celebrity endorsers, whose endorsement activities in support of a brand are often regulated to short bursts of activity fit in around other obligations. Our research is novel as it provides important insights into understanding AI influencers—a topic which, to our knowledge, has yet to be examined in the advertising literature. Specifically, our findings suggest that upon initial implementation, AI influencers are evaluated, and benefit a brand, in a manner similar to the employment of a novel celebrity endorser; thus, this research answers the call for further exploration of endorser types (Voorveld 2019), brand-building activities involving a wider variety of brand partners (Swaminathan et al. 2020), and consumer reaction to autonomous AI technologies (Schmitt 2020).

In addition, our findings show that, as with a celebrity endorser, an AI influencer who commits a transgression negatively affects attitudes and intentions toward the brand. The brand's response to the transgression, however, needs to be carefully considered. Both the pilot study and Study 2 demonstrate that AI influencers are more likely than celebrity endorsers to be viewed as similar and interchangeable and, thus, are also seen as more likely to commit comparable transgressions in the future. Therefore, to prevent damage to a brand, of the options explored, replacement of an AI influencer with a celebrity endorser is the best solution; replacement with a new AI influencer is not sufficient. Thus, this research also provides insight into strategies to effectively manage endorser transgressions (Carrillat and Ilicic 2019; Keel and Nataraajan 2012) and mitigate blame and manage crises caused by external stakeholders (Swaminathan et al. 2020).

Implications and Future Research Directions

Theoretically, much of the extant research on influencer advertising examines disclosure language (e.g., Boerman, Willemsen, and Van Der Aa 2017; Evans et al. 2017) and the factors that makes one an influencer (De Veirman, Cauberghe, and Hudders 2017; Himelboim and Golan 2019) yet fails to determine if an influencer is more or less effective than other types of endorsers (for an exception, see Schouten, Janssen, and Verspaget 2020). Our research fills this gap by directly comparing celebrity endorsers to AI influencers, a novel type of endorser. As such, we begin the investigation of how AI and nonhuman, but humanlike, agents can be influential in branding activities, thus answering the call of Schmitt (2020) and Swaminathan and colleagues (2020) to examine the impact of a wider variety of brand partners resulting from a hyperconnected world.

This research also broadens our knowledge on how consumers conceptualize the relationship between

brands and their endorsers, extending this relationship to include AI influencers. Much of the research done on AI in an online context concerns the spreading of misinformation through social bots (i.e., Broniatowski et al. 2018; Ferrara et al. 2016; Shao et al. 2018). However, our results extend this research and instead explore when utilization of AI in an online context can potentially be beneficial. Our results suggest similarities between celebrity endorsers and AI influencers, building on existing research on the positive effects of endorsers (Bergkvist and Zhou 2016; Erdogan 1999; Keel and Nataraajan 2012). Results suggest that AI influencers are perceived positively by consumers and thus provide meaningful brand benefits. Similar to celebrities, though, brands that employ AI influencers also experience the detrimental effects of endorser transgressions (Bartz, Molchanov, and Stork 2013; Fong and Wyer 2012; Louie, Kulik, and Jacobson 2001; Till and Shimp 1998; Um 2013). However, overcoming an AI influencer transgression requires a repentance strategy that employs a new, non-AI influencer. Theoretically, this aligns with existing research on aversion to various forms of technology (Dietvorst, Simmons, and Massey 2015; Venkatesh and Davis 2000; Wirtz et al. 2018) and extends research on AI into marketing and specifically the context of brand endorsers.

Managerially, this research begins to both answer questions for marketers who employ AI influencers and to raise additional questions. Our research demonstrates that an AI influencer provides the same positive brand benefits as a celebrity endorser, and Appel and colleagues (2020) suggest that AI influencers may even provide additional control and cost savings. Thus, for many brands, an AI influencer may be an effective alternative to hiring a celebrity endorser. These benefits may be especially important for smaller brands with fewer resources. Further, our research finds that brands suffer when an endorser commits a transgression, regardless of the type of endorser utilized. Thus, a celebrity does not provide an advantage in this context.

However, employing an AI influencer may not be without issue. Research already suggests that implementing AI can be challenging. In addition to the time and cost to develop AI capabilities, there are potential issues in terms of compatibility with data sources (Paschen, Pitt, and Kietzmann 2020), availability of high-quality training data, and consumer privacy concerns (Campbell et al. 2020). Research suggests that consumers are vulnerable to manipulation by bots who amplify content through strategic use of replies and mentions (Shao et al. 2018). Moreover, much of the popular press coverage on the use of AI in an online social context concerns the deceptive acts of social bots and the spread of unverified or malicious rumors that "mislead, exploit, and manipulate social media discourse" (Ferrara et al. 2016, p. 98). Thus, marketers may face a challenge of overcoming negative preconceived attitudes held by some consumers toward the use of AI.

Further, an AI influencer may confirm some of these preconceived negative notions, as the algorithms used by AI influencers to create posts and respond to followers rely on unverified inputs and, therefore, may result in the spread of misinformation. Once discovered by the public, this type of transgression may result in degradation of the brand's image in a manner similar to what happens in the aftermath of security breaches or other types of technology failures whereby brand trust, market confidence, and financial performance are damaged (Whitler and Farris 2017). Our research also suggests that if an AI influencer does commit a transgression, the marketer's options in terms of replacement may be limited. Attempts to engage in an effective repentance strategy could be further complicated if the brand has closely aligned its image with that of AI and technology. Thus, while replacing an AI influencer with a celebrity may help to temporarily attenuate reduced attitudes and purchase intentions, the overall brand positioning may suffer if the celebrity fails to provide the same type of meaning for the brand. Indeed, past research finds that revoking an association with a celebrity who has committed a transgression can negatively affect attitudes and purchase intentions if the celebrity is highly congruent with the brand (Carrillat, d'Astous, and Lazure 2013). Future research on this topic is needed not only to contribute to our theoretical understanding of different endorser types but also to further guide managerial implementation.

This research is not without limitations. The use of MTurk for data collection could be considered a limitation, although research suggests that MTurk samples result in similar findings when compared to student and other adult samples (Kees et al. 2017). Future research should investigate if the recommendations made based on the results from this study hold true with other samples. Another consideration that should be considered in future research is the transition between an original AI influencer and a replacement endorser. In this study, the AI replacement was announced via a news article. It is not known if the response would be different (or even noticed) if the

AI influencer were simply replaced without any type of announcement.

Study 1 shows that when no transgression is committed, both celebrity endorsers and AI influencers are viewed equally positively. However, both conditions considered a fictitious brand of sunglasses. It is not known if the effectiveness of an AI influencer depends on the product category being endorsed. Future research should determine if there are product categories for which either type of endorser is more or less effective. Indeed, the matchup hypothesis suggests that congruence plays an influential role in affecting brand evaluations and intentions (Kamins 1990; Till and Busler 2000; Wright 2016), and recent research on influencer advertising demonstrates that endorsed brands incongruent with an influencer's persona result in lower attitudes toward the product (Kim and Kim 2020). Innovative brands or product categories may benefit more from the use of an AI influencer as opposed to a traditional celebrity. Such product categories might include those that follow fashion trends (e.g., clothing and home furnishings) or those related to technology (e.g., electronics or automotive).

Future research should also consider if the style with which the AI influencer interacts with followers impacts the public's reaction to a transgression or a replacement of the influencer. Wu, Chen, and Dou (2017), find that consumer interactions with smart devices can either be classified as friendly, for example, reminiscent of the character Baymax in Disney movie *Big Hero 6*, or engineer-like, whereby the technology's response is precise and attentive but lacks warmth. AI influencer interaction styles can also vary in terms of level of familiarity and friendliness displayed to their followers. In the case of completely autonomous AI, the actions of the influencer will be impacted not only by the initial programing but also by the language and style of those with whom they interact. Thus, a range of interaction styles from very personable to very robotic might result. Insights into these differences might be similar to those in research related to the uncanny valley that suggests humanlike physical appearance and behavior in technology may create a feeling of unease in consumers when robots become too humanlike (e.g., Kim, Schmitt, and Thalmann 2019; Schmitt 2020).

Conversely, just as future research should examine variations in AI influencers, future research should also examine whether differences exist between AI influencers and other types of human endorsers, such as social media influencers. Past research suggests similar positive brand benefits when human celebrity endorsers and human social media influencers are employed (Bergkvist and Zhou 2016; Breves et al. 2019; Colliander and Dahlén 2011; Erdogan 1999; Keel and Nataraajan 2012; Kim and Kim 2020). However, some research suggests influencers may be more effective than celebrities under certain conditions (Schouten, Janssen, and Verspaget 2020), and recent research has started to examine more nuanced effects of celebrity endorsers when their implied brand responsibilities differ (Fowler and Thomas 2019). Thus, comparisons of AI influencers to human influencers may provide additional distinctions among endorser type, providing meaningful theoretical and managerial contributions.

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